

The birth–death–catastrophe process

Definition and main results

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Chapter 4

The birth–death–catastrophe process

4.1 Introduction

A cell can empty by ordinary deaths, or it can rupture and release its load. At the first handful of replicative cycles those fates are decided by chance: a deterministic trajectory that moves smoothly between one bacterium and none answers the wrong question. What one wants is the chance that the infection establishes itself at all, and how large the intracellular population is if it does.

The process that answers those questions is elementary. Each replicative unit inside the cell independently

- divides at rate λ ,
- dies at rate μ , or
- triggers rupture of the whole cell at rate δ .

That is the birth–death–catastrophe (BDC) process, restated as [definition 4.2](#) below. A positive integer is the intracellular load; 0 is internal extinction; H records that the cell has ruptured.

One reading is the early intracellular phase of *Y. pestis* in the macrophage: the bacterium replicates within the cell and then lyses it, releasing the accumulated load. The word “catastrophe” names that bursting event, not an adverse judgement on the bacterium, and its hazard grows with the intracellular population. Closely related stochastic models have been used for *F. tularensis* [1, 2], for *Salmonella enterica* [3], and for anthrax [4]. More broadly, the balance between establishment and collapse in small infected populations is itself a stochastic object [5, 6]. None of the formulae requires that biological reading.

The process used here is a specialisation of the birth–immigration–catastrophe model of Brockwell, Gani and Resnick [7], who wrote in 1982 that

“There has recently been a great deal of interest in the development and analysis of stochastic models for the growth of populations which are subject to catastrophes due either to death or large-scale emigrations.”

The process they introduced was more general than the one used here. It also allowed immigration, through single upward steps arriving at a constant Poisson rate, and catastrophes whose sizes were independent random variables. In the present model immigration is absent, and a catastrophe ends the intracellular process altogether rather than removing a random number of individuals. The population moves directly to 0, or to a distinct post-catastrophe state H , and thereafter remains fixed. The mathematics rests on the classical theory of birth–death processes and their killed variants [8, 9, 10].

The derivation proceeds as follows.

- A first-step argument produces an autonomous Riccati equation for the non-catastrophe probability $I(t)$.
- Internal extinction $D(t)$ obeys the same flow from a different initial condition; their difference is the productive probability $I_{\text{fix}} = I - D$.
- The mean load is $J = -I'/\delta$; higher load moments are polynomials in I .
- Release moments follow by balance, using the same higher-order terms.
- Burst size and burst time are read off from those functions.

Once I is known, no other state probability is needed to advance it in time. The roots $a > 1 > b$ of the characteristic quadratic make the closed forms readable. The lower root b is the probability of internal extinction before rupture; $1/a$ is the ratio of the geometric burst-size law; their difference supplies the time scale $\theta = \lambda(a - b)$. Write $A = a - 1$ and $B = 1 - b$.

The principal results, in advance, are these. From one founder, $I(t)$, $D(t)$ and $I_{\text{fix}}(t)$ are explicit in a , b and $w = e^{\theta t}$. The mean load and mean release are

$$J = -\frac{\lambda}{\delta}(I - a)(I - b), \quad V = (1 - I)\left[1 + \frac{\lambda}{\delta}(1 - I)\right],$$

with late-time release $V_{\infty} = aB/A$. Neither is a typical-cell history: J mixes large loads from a dwindling productive set with zeros from every path that has finished, and V_{∞} averages over realisations that never rupture. Conditional on rupture, the first two burst-size moments match a geometric law of ratio $1/a$; the law itself is derived in Chapter 5. Variances of load and release are likewise functions of I .

Remark 4.1 (What is classical here, and what is developed here). The Riccati equation for the non-catastrophe probability, together with its two roots, belongs to the classical theory of birth–death processes with killing [9]. The process is a special case of the birth–immigration–catastrophe model of Brockwell, Gani and Resnick [7], and the lower root b is the familiar extinction fixed point from branching-process theory [11, 12].

The present treatment keeps the released count W_t alongside the load X_t , which gives its first two moments and the corresponding conditional burst moments. It also keeps non-catastrophe, internal extinction and continued productivity separate; this distinction is lost when death and rupture share a single absorbing state, yet it is needed for the quasi-stationary analysis of Chapter 5. Finally, expressing the load moments as polynomials in I supplies the age-dependent quantities used by the population model of Chapter 6.

Four assumptions are load-bearing, and all four are contestable.

1. **Linear birth.** Each particle divides at rate λ whatever the current load: there is no intracellular carrying capacity.
2. **Catastrophe hazard δn .** Rupture is triggered at rate δn from load n , not δn^2 and not a threshold at a critical load.
3. **Instantaneous, complete release.** The whole load leaves the cell at one instant.
4. **Constant rates.** No eclipse phase and no age-dependence within the cell.

Section 4.2 defines the pair (X_t, W_t) and fixes the conventions. Section 4.3 derives and solves the scalar equation for the three cell-fate probabilities. The moment calculation follows in section 4.4, after which section 4.5 explains the special role of I . Section 4.6 turns the moment results into statements about burst size and burst time, and section 4.7 collects the formulae for later use. The final section records the boundary of the calculation and the connection to the next two chapters; section 4.A records the principal limiting checks.

4.2 The process

The birth–death–catastrophe process was introduced in the mathematical introduction as Definition 2.4. We restate it here for the single-cell reading.

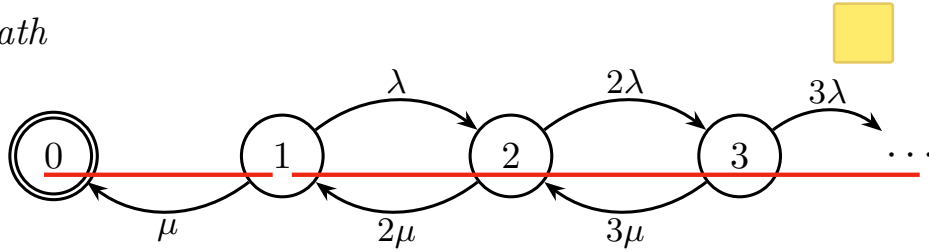
Definition 4.2 (Birth–death–catastrophe process). Let X_t take values in $\{0, 1, 2, \dots\} \cup \{H\}$ and record the intracellular load while the cell remains intact. Each replicative unit independently gives birth at rate λ , dies at rate μ , and triggers rupture at rate δ . For $n \geq 1$, the non-zero transition rates are

$$q_{n,n+1} = \lambda n, \quad q_{n,n-1} = \mu n, \quad q_{n,H} = \delta n, \quad (4.1)$$

and both 0 and H are absorbing.

The units counted by X_t are infectious particles inside a host cell, and rupture releases the entire load at once. Because the total catastrophe rate in state n is δn , a crowded cell is more likely to burst than a sparsely occupied one. This population dependence is the small addition that changes the character of the process. The same chain is drawn beside the linear birth–death process in Figure 2.6 of the mathematical introduction; figure 4.1 repeats it here so that the two absorbing mechanisms remain in view.

birth–death



birth–death–catastrophe

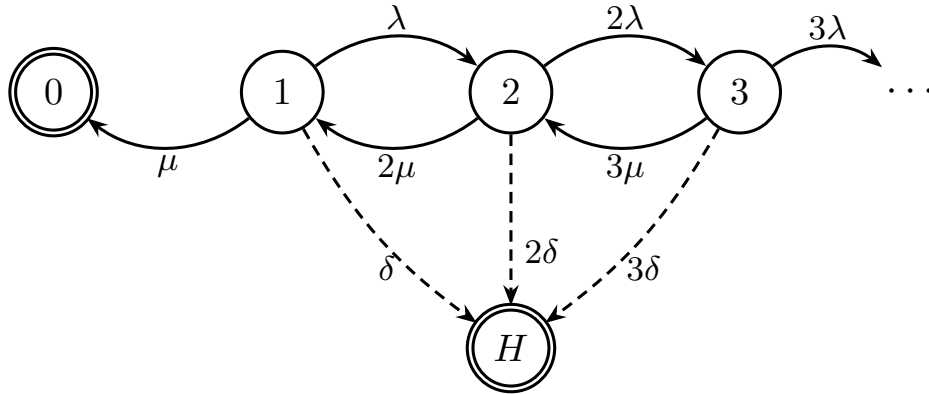


Figure 4.1: Linear birth–death dynamics (top) and the birth–death–catastrophe process used throughout this chapter (bottom). Double circles are absorbing. Ordinary extinction enters 0 only from state 1; catastrophe jumps from any positive count to the distinct state H , at total rate δn .

4.2.1 Sample paths

Unless a caption states otherwise, figures use the working point $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$, with $X_0 = 1$.

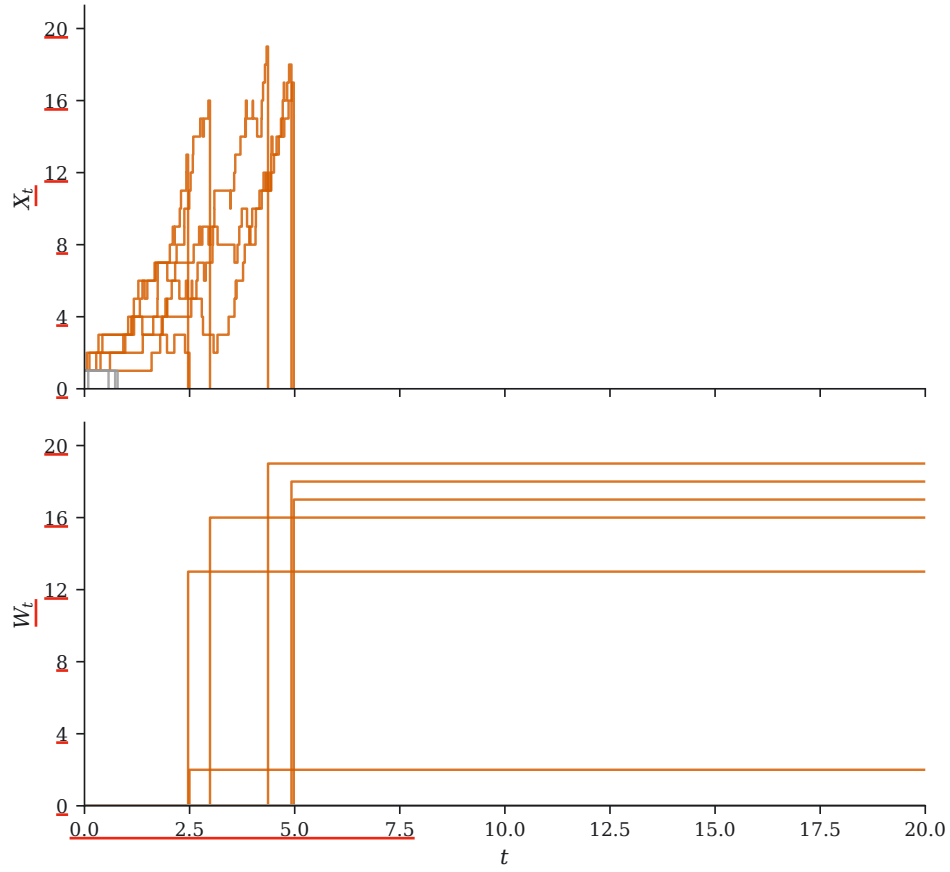


Figure 4.2: Ten exact realisations of the load X_t (top) and the released count W_t (bottom), generated by the Gillespie algorithm [13]. The step of X_t to the plotting baseline marks entry into H , not the count state 0. Each W_t path remains at zero until rupture, then jumps to the load present at that instant. $\lambda = 1$, $\mu = 0.2$, $\delta = 0.05$, $X_0 = 1$; seed 12.

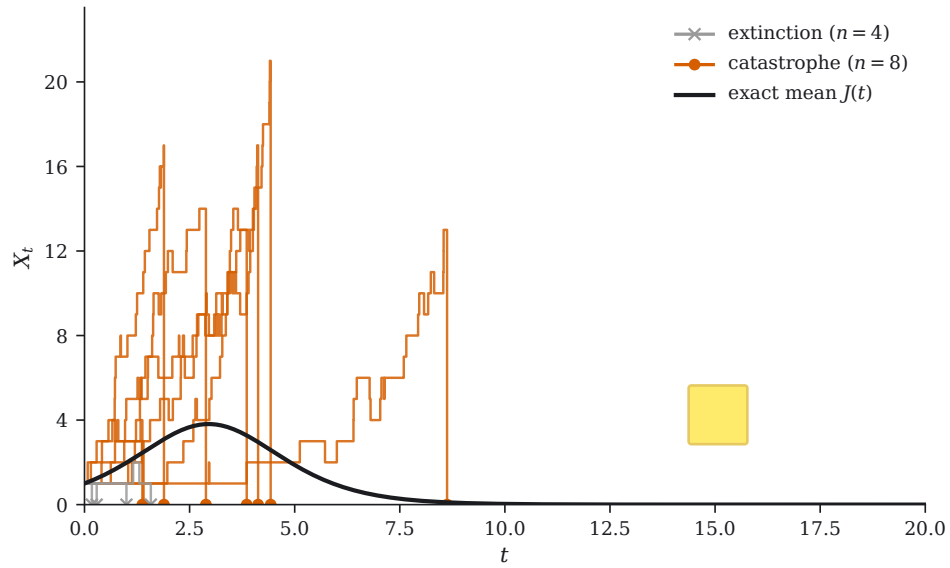


Figure 4.3: Twelve exact realisations of X_t , coloured by terminal fate. Grey paths ending in crosses reach internal extinction; vermilion paths ending in filled circles enter H . The bold curve is the exact mean $J(t)$. $\lambda = 1$, $\mu = 0.2$, $\delta = 0.05$, $X_0 = 1$; seed 154.

Figure 4.2 shows a handful of exact realisations of the pair (X_t, W_t) . Some paths decline to 0; others grow for a time and end abruptly at H . The corresponding release paths contain only one possible movement: a jump from zero to the load carried at rupture.

Colouring the paths by their eventual fate, as in figure 4.3, makes visible what a single undifferentiated bundle conceals. The exact mean $J(t) = \mathbb{E}(X_t)$ passes through the ensemble, but it belongs to neither class of path.

Remark 4.3 (The mean load describes no typical path). After a realisation reaches 0 or H , its numerical load is zero. The late-time mean therefore averages substantial loads from a shrinking set of productive paths against zeros from every path that has finished. It is neither the load of a typical cell nor the load of a typical surviving cell. The latter is the conditional mean J/I_{fix} , considered in section 4.6.3.

4.2.2 The released count and the joint process

Rupture creates a second count, the number of particles released into the extracellular medium. It remains zero while the cell is intact and, in the absence of any imposed extracellular dynamics, remains fixed after release. Let

$$\tau = \inf\{t : X_t = H\}$$

be the catastrophe time. The released count is

$$W_t = \begin{cases} 0, & t < \tau, \\ X_{\tau-}, & t \geq \tau. \end{cases} \quad (4.2)$$

The burst size is the load present an instant before rupture, $\mathcal{K} := X_{\tau-}$, as recorded in equation 2.20 of the mathematical introduction. The left limit matters: at τ the internal state has become H , while the released load is the integer present an instant earlier.

The joint transition list makes the relation between the two counts explicit. Writing $\Delta X_t = X_{t+\Delta t} - X_t$ and similarly for W_t , and conditioning on $X_t = n \geq 1$,

$$\begin{aligned} \mathbb{P}(\Delta X_t = 1, \Delta W_t = 0 \mid X_t = n) &= \lambda n \Delta t + o(\Delta t) && \text{(birth)} \\ \mathbb{P}(\Delta X_t = -1, \Delta W_t = 0 \mid X_t = n) &= \mu n \Delta t + o(\Delta t) && \text{(death)} \\ \mathbb{P}(X_{t+\Delta t} = H, W_{t+\Delta t} = n \mid X_t = n) &= \delta n \Delta t + o(\Delta t) && \text{(catastrophe)} \\ \mathbb{P}(\Delta X_t = 0, \Delta W_t = 0 \mid X_t = n) &= 1 - (\lambda + \mu + \delta)n \Delta t + o(\Delta t) && \text{(no event)}. \end{aligned} \quad (4.3)$$

This pair (X_t, W_t) will be followed through its means, variances and covariance. Most of the work lies in the marginal moments; the joint structure, as we shall see, is unusually simple.

4.2.3 Two conventions for rupture

There is another reasonable way to represent rupture: send the process to 0 rather than to a separate state H . After all, once a cell has ruptured it contains no replicative units, although the cell itself has also ceased to exist. We call the transition to 0 *killing* and the transition to H *catastrophe*. Figure 4.4 shows the distinction.

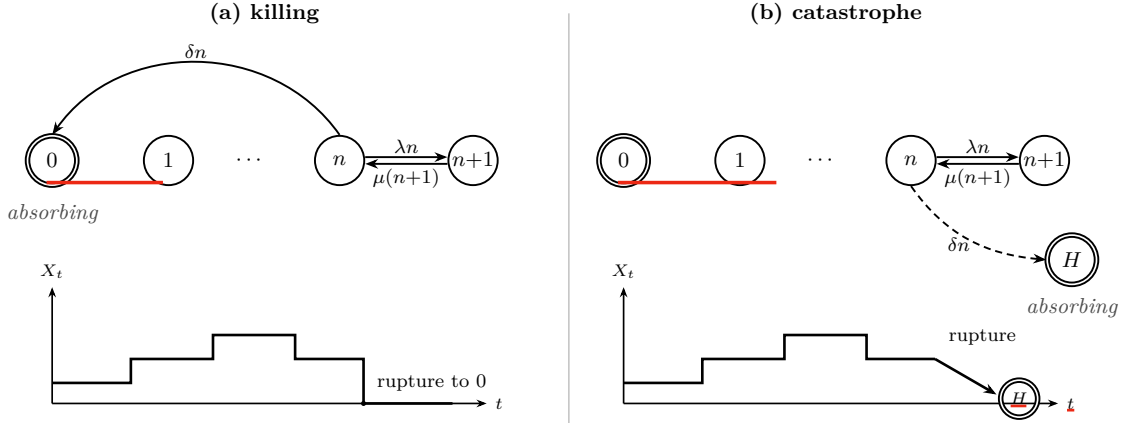


Figure 4.4: Two conventions for rupture. Under killing (left), rupture from state n ends at 0 and is indistinguishable from internal extinction. Under catastrophe (right), used here, the same transition ends at the separate state H . Birth and death rates remain λn and μn .

The choice does not alter the observed burst times or burst sizes. It alters what can be distinguished after absorption. Under killing, internal extinction and rupture arrive at the same state; their probabilities cannot be recovered separately from the state occupancy. With H kept apart, we may distinguish the internal-extinction probability D , the rupture probability $1 - I$, and the productive probability $I_{\text{fix}} = I - D$. That distinction is the reason for using H throughout. Chapter 5 carries both conventions through the generating-function treatment.

4.2.4 Notation

Although H is a label rather than a number, after rupture the intracellular *load* is zero. Whenever X_t appears arithmetically—in a sum, product or moment—we use the numerical-load convention

$$X_t = n \mapsto n, \quad X_t = H \mapsto 0.$$

Thus H remains distinct from 0 as a state, allowing the two absorbing mechanisms to be distinguished, but contributes zero to $\mathbb{E}(X_t^p)$ and to products such as $X_t W_t$.

One possible collision has been removed now: $\mathcal{K}$ denotes the burst size, whereas $K(t) = \mathbb{E}(X_t^2)$ will denote the second moment of the load. They are unrelated. The full list of symbols appears in table 4.1. Unless explicitly stated otherwise, the process begins with one unit, $X_0 = 1$.

4.3 The Riccati equation and its two roots

Three probabilities sit close together in this process and are easily confused: the cell may have avoided rupture, it may have died internally, or it may still be productive. Their meanings differ, yet all three can be obtained from one scalar differential equation. We derive that equation first, then follow its flow from two initial conditions.

4.3.1 First-step decomposition

Let

$$I(t) = \mathbb{P}(X_t \neq H \mid X_0 = 1)$$

be the *non-catastrophe probability*, the chance that rupture has not occurred by time t . A path counted by $I(t)$ may still be productive or it may already have died at 0; it is excluded only if it has reached H .

Consider the first event experienced by the founding unit during a short interval. Time homogeneity allows the process left after that event to be followed for a further duration t :

$$I(t + \Delta t) = \mu \Delta t + [1 - (\lambda + \mu + \delta)\Delta t]I(t) + \lambda \Delta t I(t)^2 + o(\Delta t).$$

If the founder dies, rupture has been avoided for good, giving the first term. A rupture contributes nothing. If the founder divides, both descendant families must avoid rupture, and the branching property gives probability $I(t)^2$. After rearrangement,

$$\frac{I(t + \Delta t) - I(t)}{\Delta t} = \mu - (\lambda + \mu + \delta)I(t) + \lambda I(t)^2 + \frac{o(\Delta t)}{\Delta t}.$$

Taking $\Delta t \rightarrow 0$ gives

$$\frac{dI}{dt} = \mu - (\lambda + \mu + \delta)I + \lambda I^2, \quad I(0) = 1. \quad (4.4)$$

This is a Riccati equation, but its useful feature is simpler than the name suggests: the right-hand side contains I and the three rates, and nothing else. The equation closes before any moments have been introduced.

4.3.2 The two roots

The quadratic on the right of (4.4) becomes transparent once its roots are named.

Lemma 4.4 (The characteristic quadratic has one root on either side of unity). *Let $\lambda > 0$, $\mu \geq 0$ and $\delta > 0$, and set*

$$q(x) = \lambda x^2 - (\lambda + \mu + \delta)x + \mu, \quad \eta = \frac{\lambda + \mu + \delta}{2\lambda}.$$

Then q has two real roots

$$a = \eta + \sqrt{\eta^2 - \frac{\mu}{\lambda}}, \quad b = \eta - \sqrt{\eta^2 - \frac{\mu}{\lambda}},$$

satisfying $b < 1 < a$, with $ab = \mu/\lambda$ and $a + b = (\lambda + \mu + \delta)/\lambda$. Moreover $b \in [0, 1)$, with $b = 0$ if and only if $\mu = 0$.

Proof. The roots are real and distinct because

$$\eta^2 - \frac{\mu}{\lambda} = \frac{(\lambda + \mu + \delta)^2 - 4\lambda\mu}{4\lambda^2} = \frac{(\lambda - \mu)^2 + \delta(\delta + 2\lambda + 2\mu)}{4\lambda^2} > 0$$

when $\delta > 0$. Since $q(1) = \lambda - (\lambda + \mu + \delta) + \mu = -\delta < 0$, the value at 1 lies below the axis of an upward-opening parabola, so one root lies on each side of unity. Finally, Vieta's relation gives $ab = \mu/\lambda \geq 0$. As $a > 0$, this implies $b \geq 0$, with equality precisely when $\mu = 0$. $\square$

Remark 4.5 (Standing assumptions). Unless a limiting case is being considered, we take $\lambda > 0$, $\mu \geq 0$ and $\delta > 0$, and set $X_0 = 1$. The case $\delta = 0$ is the linear birth–death process of Section 2.2.3 of the mathematical introduction; it returns in [section 4.A.4](#) as a limit. The linear state dependence of the rates also makes the process non-explosive, with finite moments at every finite time. This permits the differentiations under expectation used in [section 4.4](#).

The Riccati equation now reads

$$\frac{dI}{dt} = \lambda(I - a)(I - b). \quad (4.5)$$

This factorisation gives a phase-line picture before it gives a formula. The initial value $I(0) = 1$ lies between b and a , where the derivative is negative. Consequently $I(t)$ decreases towards b

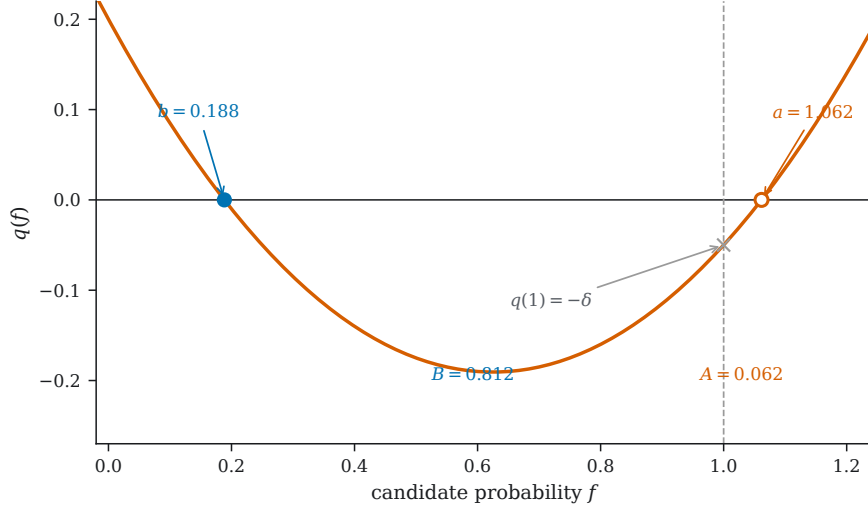


Figure 4.5: The characteristic quadratic and its flow for $\lambda = 1$, $\mu = 0.2$ and $\delta = 0.05$. The roots $b = 0.188$ and $a = 1.062$ lie on either side of unity because $q(1) = -\delta$. A trajectory beginning at $I(0) = 1$ moves towards the stable root b . The distances $A = a - 1$ and $B = 1 - b$ recur in the closed forms.

and never crosses it. Figure 4.5 draws this motion; the single observation $q(1) = -\delta$ places the initial point between the two roots.

Two short distances from unity make the later algebra easier to read:

$$A = a - 1, \quad B = 1 - b.$$

Vieta's relations then yield

$$AB = (a - 1)(1 - b) = a + b - ab - 1 = \frac{\lambda + \mu + \delta}{\lambda} - \frac{\mu}{\lambda} - 1 = \frac{\delta}{\lambda},$$

while $A + B = a - b$. We collect the identities here and name the exponential rate that will recur:

$$ab = \frac{\mu}{\lambda}, \quad a + b = \frac{\lambda + \mu + \delta}{\lambda}, \quad AB = \frac{\delta}{\lambda}, \quad A + B = a - b, \quad \theta = \lambda(a - b). \quad (4.6)$$

The point of these relations is practical. Expressions which are cumbersome in λ , μ and δ often fall into a recognisable shape in a , b , A and B .

4.3.3 Solving for $I(t)$

Equation (4.5) separates:

$$\int \frac{dI}{(I - a)(I - b)} = \lambda t + \text{const.}$$

Using

$$\frac{1}{(I - a)(I - b)} = \frac{1}{a - b} \left(\frac{1}{I - a} - \frac{1}{I - b} \right),$$

we obtain

$$\frac{I(t) - a}{I(t) - b} = \frac{I(0) - a}{I(0) - b} e^{\lambda(a-b)t} = \frac{I(0) - a}{I(0) - b} w, \quad w := e^{\theta t}. \quad (4.7)$$

For $I(0) = 1$, the initial ratio is $-A/B$, and solving for $I(t)$ gives the first main formula.

Theorem 4.6 (The non-catastrophe probability). *From one founder, the probability that catastrophe has not occurred by time t is*

$$I(t) = \frac{aB + bAw}{B + Aw}, \quad w = e^{\theta t}, \quad \theta = \lambda(a - b), \quad (4.8)$$

where a and b are given in [lemma 4.4](#), with $A = a - 1$ and $B = 1 - b$.

As t grows, w grows and

$$\lim_{t \rightarrow \infty} I(t) = \lim_{t \rightarrow \infty} \frac{aB + bAw}{B + Aw} = b.$$

When $\mu > 0$, the paths which avoid rupture indefinitely are exactly those which die internally first. Thus b is the probability of internal extinction before rupture. When $\mu = 0$, death is impossible, $b = 0$, and every path eventually ruptures. The lower root is therefore both the stable point of the differential equation and a probability. The upper root is not visited by $I(t)$, although it will acquire its own probabilistic role through the burst-size law.

4.3.4 The same equation again: internal extinction

Let

$$D(t) = p_0(t) = \mathbb{P}(X_t = 0 \mid X_0 = 1)$$

be the probability of internal extinction by time t . The first-step argument has the same shape as before. Death of the founder contributes certainty; division requires both descendant families to die; rupture contributes nothing. Consequently

$$\frac{dD}{dt} = \mu - (\lambda + \mu + \delta)D + \lambda D^2, \quad D(0) = 0. \quad (4.9)$$

The meanings of I and D differ, but their motion is governed by the same flow. For $b > 0$, inserting $D(0) = 0$ into (4.7) gives the initial ratio a/b and hence the following expression. When $b = 0$, internal extinction is impossible and the same formula is recovered directly as $D \equiv 0$.

Theorem 4.7 (The internal-extinction probability). *The probability that the process has died out internally by time t is*

$$D(t) = \frac{ab(w - 1)}{aw - b}. \quad (4.10)$$

The two solutions approach b from opposite sides: I descends from 1 and D rises from 0. Their narrowing separation is the probability that the cell is still productive.

4.3.5 The non-fixation probability

Define

$$I_{\text{fix}}(t) = \mathbb{P}(X_t \notin \{0, H\} \mid X_0 = 1).$$

Since 0 and H are the only ways to leave the productive states,

$$I_{\text{fix}}(t) = I(t) - D(t).$$

Equivalently, if T_0 and T_H are the two hitting times, then $I_{\text{fix}}(t) = \mathbb{P}(t < T_0 \wedge T_H)$. This is the survival event used for conditioning in Chapter 5: the cell must remain intact *and* retain a positive load.

A direct first-step calculation is useful because it anticipates the multi-founder case. Following a birth, the combined process is productive at time t provided no descendant family has triggered

catastrophe and the two families have not both died out. Those events have probabilities $I(t)^2$ and $D(t)^2$, respectively, so the birth branch contributes $I^2 - D^2$. Thus

$$\frac{dI_{\text{fix}}}{dt} = -(\lambda + \mu + \delta)I_{\text{fix}} + \lambda(I^2 - D^2), \quad I_{\text{fix}}(0) = 1. \quad (4.11)$$

Subtracting (4.9) from (4.4) gives the same equation, as it should.

Theorem 4.8 (The non-fixation probability). *The probability that the process remains productive at time t is*

$$I_{\text{fix}}(t) = I(t) - D(t) = \frac{aB + bAw}{B + Aw} - \frac{ab(w-1)}{aw-b} = \frac{(a-b)^2 w}{(B + Aw)(aw-b)}. \quad (4.12)$$

The boundary values fit the interpretation. At $t = 0$, the last expression is one; as $t \rightarrow \infty$, its numerator grows as w and its denominator as w^2 , so it tends to zero. Evaluating the differential equation at the origin gives

$$I'_{\text{fix}}(0) = -(\lambda + \mu + \delta) + \lambda = -(\mu + \delta),$$

the combined rate at which the founding unit is lost either to death or to rupture. A birth does not end productivity.

If $\mu = 0$, then $b = 0$ and $D \equiv 0$, so $I_{\text{fix}} = I$. When $\mu > 0$, the two functions separate, and conditioning on non-catastrophe alone admits cells which are intact but already empty. Figure 4.6 shows this separation and the eventual closing of the gap.

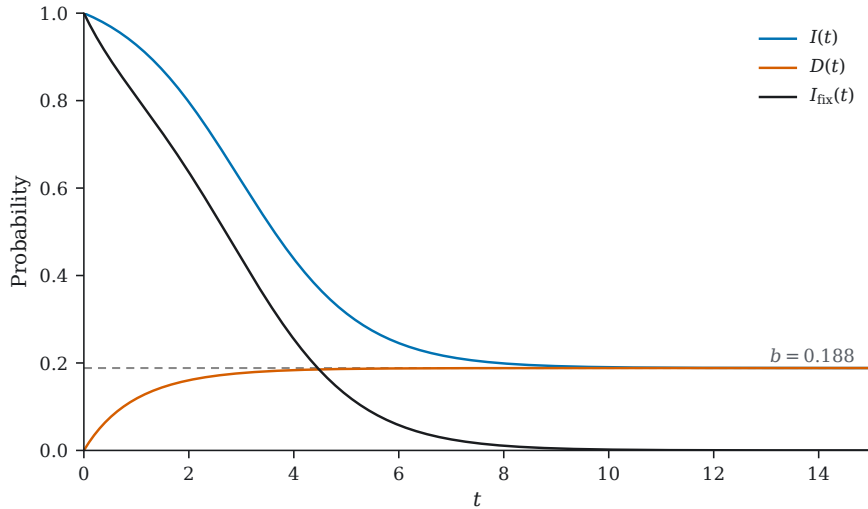


Figure 4.6: Non-catastrophe, internal-extinction, and productive probabilities for $\lambda = 1$, $\mu = 0.2$ and $\delta = 0.05$. The non-catastrophe probability $I(t)$ falls to b while the internal-extinction probability $D(t)$ rises to it. Their difference $I_{\text{fix}}(t)$ is the probability of continued productivity and tends to zero.

4.3.6 Several founders

Now suppose that the cell begins with $X_0 = k$. Up to the first catastrophe, the family trees and trigger clocks descending from the founders may be constructed independently, as described in Section 2.2.4 of the mathematical introduction. The chance that none of the k families has triggered rupture by time t is

$$I_k(t) = \mathbb{P}(X_t \neq H \mid X_0 = k) = I(t)^k,$$

whereas extinction of the whole population requires every family to die, giving $D_k(t) = D(t)^k$. Their difference is

$$I_{\text{fix},k}(t) = \mathbb{P}(X_t \notin \{0, H\} \mid X_0 = k) = I(t)^k - D(t)^k. \quad (4.13)$$

Remark 4.9 (Non-fixation does not factorise). For $\mu > 0$, the tempting identity $I_{\text{fix},k} = I_{\text{fix}}^k$ is false. If one family dies while another remains productive, the cell has not fixed; these mixed configurations are retained by $I^k - D^k$ but omitted by $(I - D)^k$. Catastrophe is different from the death of one lineage: the first trigger ruptures the cell and stops all lineages at once. Only when $\mu = 0$, so that $D \equiv 0$, do the two expressions coincide.

Figure 4.7 makes the omitted configurations visible. For two founders the naive power lies below the exact probability, and the gap is the probability that one lineage has died while the other is still productive.

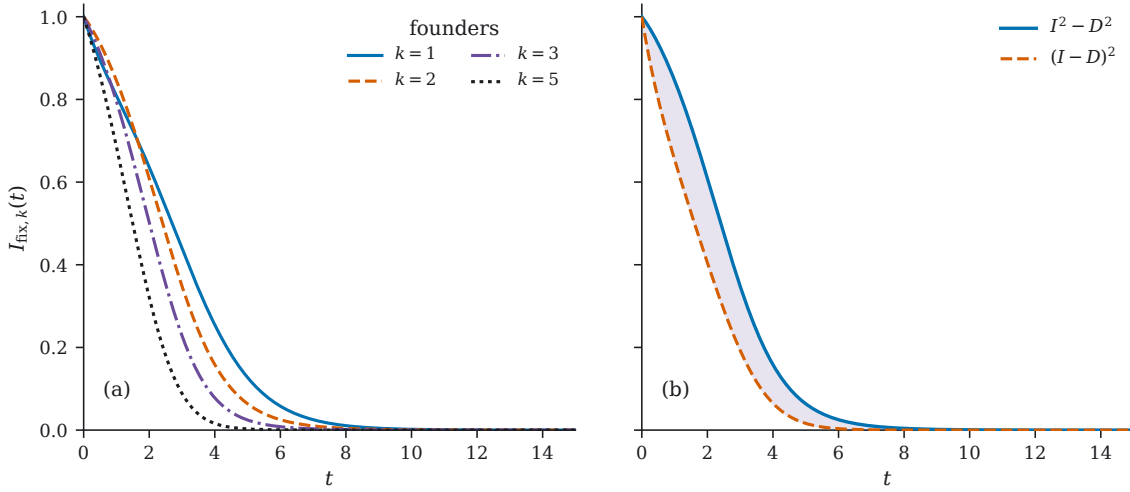


Figure 4.7: Non-fixation from several founders. (a) Exact probabilities $I_{\text{fix},k}(t) = I(t)^k - D(t)^k$ for $k = 1, 2, 3, 5$. (b) For two founders, $[I(t) - D(t)]^2$ lies below the exact result because it omits mixed configurations with one extinct and one productive lineage. $\lambda = 1$, $\mu = 0.2$ and $\delta = 0.05$.

These multi-founder quantities return in Chapter 5, where the generating functions provide a second derivation and allow several incoming particles to found one infection.

4.4 Moments

We now turn from the fates of the cell to the numbers it carries and releases. At first sight their moments appear awkward. Since the catastrophe rate grows with the population, the equation for each load moment calls for the moment above it. One would normally expect an open hierarchy and, sooner or later, a closure approximation. The Riccati equation offers a way around this.

Remark 4.10 (Why closed forms exist). Let $J(t) = \mathbb{E}(X_t)$ and $K(t) = \mathbb{E}(X_t^2)$. Their moment equations begin

$$\dot{J} = (\lambda - \mu)J - \delta K, \quad \dot{K} = 2(\lambda - \mu)K + (\lambda + \mu)J - \delta \mathbb{E}(X_t^3),$$

and the pattern continues. The hierarchy is genuinely open. However, the non-catastrophe probability obeys the autonomous equation (4.5), while $J = -I'/\delta$. Differentiating that scalar equation produces K in terms of I and J , without introducing the third moment. Repeated differentiation does the same at higher orders. Thus the open hierarchy remains present, but it

need not be truncated; it can be approached from the closed equation for I . [Proposition 4.17](#) later states the general result.

The calculation has a natural order. We first recover the mean load from I , then use its derivative to obtain the second moment. The release moments follow from balance equations in which the same higher-order terms appear and can be eliminated.

4.4.1 The mean load

The link between I and the load is especially direct. Since H is entered from state n at rate δn , the forward equation for the catastrophe state is

$$\frac{d}{dt}\mathbb{P}(X_t = H) = \delta \sum_{n \geq 1} n p_n(t) = \delta J(t).$$

Because $I(t) = 1 - \mathbb{P}(X_t = H)$,

$$\frac{dI}{dt} = -\delta J. \quad (4.14)$$

The slope of the non-catastrophe curve is therefore the negative mean load, scaled by the per-unit catastrophe rate. Differentiating [\(4.8\)](#), or substituting the Riccati equation into [\(4.14\)](#), gives the closed form.

Theorem 4.11 (The mean load). *The expected number of replicative units present at time t is*

$$J(t) = \mathbb{E}(X_t) = \frac{(a-b)^2 w}{(B + Aw)^2} = -\frac{\lambda}{\delta}(I-a)(I-b). \quad (4.15)$$

The second expression will be the more useful one: it places J directly on the Riccati phase line. From the initial load of one, the mean may first rise as productive paths grow, but it eventually returns to zero as absorption becomes certain. This fall does not mean that the remaining productive cells are empty. It means that there are progressively fewer of them, as [remark 4.3](#) made clear.

To expose the second moment, consider the expected change of X_t over Δt . A birth adds one at rate λX_t , a death removes one at rate μX_t , and a catastrophe removes the whole numerical load at rate δX_t . Hence

$$\mathbb{E}(X_{t+\Delta t}) = \mathbb{E}(X_t) + \mathbb{E}\left(\lambda \Delta t X_t - \mu \Delta t X_t - \delta \Delta t X_t^2\right) + o(\Delta t),$$

and therefore

$$\frac{dJ}{dt} = (\lambda - \mu)J - \delta K(t), \quad K(t) := \mathbb{E}(X_t^2). \quad (4.16)$$

The term $-\delta K$ is the distinctive one. A large load is both costly when removed and more likely to be removed, so its contribution is quadratic.

4.4.2 The second moment

Although [\(4.16\)](#) couples J to K , it can be read in reverse because J is already known. Differentiate [\(4.14\)](#):

$$\frac{dJ}{dt} = -\frac{1}{\delta} \frac{d^2 I}{dt^2}. \quad (4.17)$$

Differentiating the Riccati equation gives

$$\frac{d^2 I}{dt^2} = [-(\lambda + \mu + \delta) + 2\lambda I] \frac{dI}{dt},$$

and using $I' = -\delta J$ once more,

$$\frac{d^2 I}{dt^2} = (\lambda + \mu + \delta)\delta J - 2\lambda\delta IJ. \quad (4.18)$$

Equations (4.17) and (4.18) imply

$$\frac{dJ}{dt} = -(\lambda + \mu + \delta)J + 2\lambda IJ.$$

Comparison with (4.16) then gives

$$(\lambda - \mu)J - \delta K = -(\lambda + \mu + \delta)J + 2\lambda IJ,$$

from which K follows without ever computing $\mathbb{E}(X_t^3)$.

Theorem 4.12 (The second moment of the load). *Let $\kappa = 1 + \delta/(2\lambda)$. Then*

$$K(t) = \mathbb{E}(X_t^2) = \left[1 + \frac{2\lambda}{\delta}(1 - I)\right]J = \frac{2\lambda}{\delta}(\kappa - I)J, \quad (4.19)$$

and, after substituting (4.15),

$$K(I) = \frac{2\lambda^2}{\delta^2}(I - \kappa)(I - a)(I - b). \quad (4.20)$$

As a polynomial in I , the second moment shares the roots a and b with the mean. Its third root κ lies just above unity, outside the interval $(b, 1]$ traced by the process.

4.4.3 The mean release

The released count W_t remains at zero until rupture and then takes the load present at that instant. A rupture from state n increases W_t by n at rate δn , and so

$$\frac{dV}{dt} = \delta K(t), \quad V(t) := \mathbb{E}(W_t). \quad (4.21)$$

Release is driven by the second moment rather than the first: fuller cells release more and are also more liable to rupture.

Combining (4.21) with (4.16) eliminates K :

$$\frac{dJ}{dt} = (\lambda - \mu)J - \frac{dV}{dt}.$$

Since $J = -I'/\delta$,

$$\frac{dV}{dt} = -\frac{\lambda - \mu}{\delta} \frac{dI}{dt} - \frac{dJ}{dt},$$

which can be integrated immediately.

Theorem 4.13 (The mean release). *With $V(0) = 0$ and $J(0) = I(0) = 1$,*

$$V(t) = \mathbb{E}(W_t) = \frac{\lambda - \mu}{\delta}[1 - I(t)] - J(t) + 1 = (1 - I)\left[1 + \frac{\lambda}{\delta}(1 - I)\right]. \quad (4.22)$$

Its late-time value is

$$V_\infty = \lim_{t \rightarrow \infty} V(t) = \frac{\lambda - \mu}{\delta}(1 - b) + 1 = \frac{aB}{A}. \quad (4.23)$$

When $\mu = 0$, this reduces to $V_\infty = 1 + \lambda/\delta$.

The form in I follows by substituting (4.15) into the first expression and using (4.6). Notice that V_∞ averages over every realisation. Paths which die internally contribute zero, so it is not yet the mean conditional burst size; that distinction is taken up in [section 4.6.1](#).

4.4.4 Load and release together

Figure 4.8 restores the two familiar time courses alongside the parametric view. In time, J rises and falls while V accumulates. Eliminating time shows the same movement as one trajectory in the (J, V) plane. The two panels are complementary: the first makes the timing legible, and the second shows how intracellular load is exchanged for expected release.

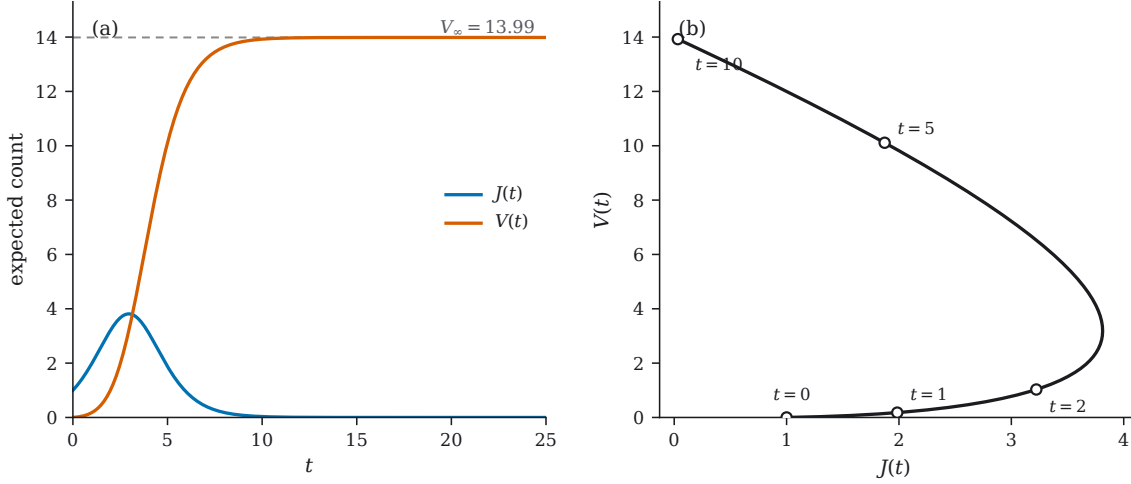


Figure 4.8: Mean intracellular load and release for $\lambda = 1$, $\mu = 0.2$ and $\delta = 0.05$. (a) The time courses $J(t)$ and $V(t)$: the load first grows and then disappears, while release increases to $V_\infty = 13.99$. (b) The same solution as the parametric path $(J(t), V(t))$, with selected times marked.

The second moment of the release is obtained by the same elimination: a rupture from state n raises W_t^2 from zero to n^2 at rate δn , so $\mathbb{E}(W_t^2)$ is driven by the third load moment, which is already present in the equation for K . The calculation is in [section 4.A.7](#).

Theorem 4.14 (The second moment of the release). *The second moment of the released count is*

$$\mathbb{E}(W_t^2) = \frac{2(\lambda - \mu)}{\delta} V - \frac{\lambda + \mu}{\delta} I - K + \frac{\lambda + \mu}{\delta} + 1. \quad (4.24)$$

4.4.5 Variances

The variances now require only $\text{Var}(Y) = \mathbb{E}(Y^2) - \mathbb{E}(Y)^2$. For the load,

$$\text{Var}(X_t) = K(t) - J(t)^2 = \frac{2\lambda}{\delta} (\kappa - I) J - J^2, \quad (4.25)$$

and for the release,

$$\text{Var}(W_t) = \frac{2(\lambda - \mu)}{\delta} V - \frac{\lambda + \mu}{\delta} I - K + \frac{\lambda + \mu}{\delta} + 1 - V^2. \quad (4.26)$$

Since J , K and V are functions of I , the two variances are as well.

Figure 4.9 shows why these expressions are worth carrying. The spread of the intracellular load soon exceeds its mean. At late times, variation in W_t reflects a mixture of paths that never ruptured and paths that released a sometimes large load. Neither panel is well described by its central curve alone.

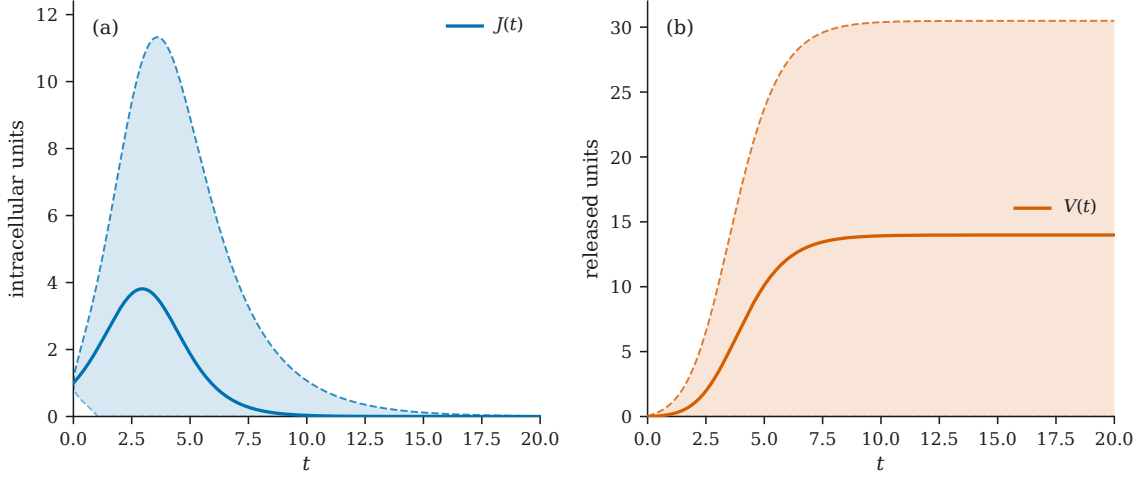


Figure 4.9: Means and one-standard-deviation envelopes for (a) the intracellular load X_t and (b) the released count W_t , with $\lambda = 1$, $\mu = 0.2$ and $\delta = 0.05$. Lower envelopes are clipped at the support boundary zero.

4.4.6 The pair (X_t, W_t)

The two components have now been treated separately. Their joint behaviour has a useful degeneracy: internal load and released load cannot be positive at the same time.

Corollary 4.15 (Exclusion, and the covariance). *At every t , $X_t W_t \equiv 0$ almost surely under the numerical-load convention of [section 4.2.4](#). Before rupture $W_t = 0$, and after rupture the state is H with numerical load zero. Hence $\mathbb{E}(X_t W_t) = 0$ and*

$$\text{Cov}(X_t, W_t) = -J(t)V(t) \leq 0. \quad (4.27)$$

The sum $X_t + W_t$ has a similarly plain interpretation. It counts particles currently held inside the intact cell together with particles already released. It does not count particles lost through internal death, and it is not a cumulative count of all births.

Corollary 4.16 (Current internal and released yield). *The mean number of particles either still inside or already released is*

$$\mathbb{E}(X_t + W_t) = 1 + \frac{\lambda - \mu}{\delta} [1 - I(t)]. \quad (4.28)$$

Algebraically, (4.28) is a rearrangement of (4.22). Probabilistically, it describes the surviving inventory of the process: the founder, together with the net contribution of births and deaths accumulated before time t or rupture. This is the quantity whose late-time limit underlies V_∞ .

4.5 The I -economy

The formulae of the previous section share a curious economy: although the process evolves in time, each moment can be written as a function of I alone. This is more than a change of variable. It says that the stochastic dynamics move along a family of algebraic curves already fixed by the Riccati equation.

On the interval (b, a) , the right-hand side of (4.5) is negative. Beginning at $I(0) = 1$, the function $I(t)$ therefore decreases strictly towards b , giving a bijection from $[0, \infty)$ to $(b, 1]$. It may be used as a clock, albeit one which runs backwards: once a quantity is known as a function

of I , its dependence on time is determined by the speed

$$\frac{dI}{dt} = \lambda(I - a)(I - b).$$

The first two load moments suggest that the polynomial structure continues, and indeed it does.

Proposition 4.17 (Every load moment is a polynomial in I). *For each $p \geq 1$, there is a polynomial Q_p such that*

$$\mathbb{E}(X_t^p) = Q_p(I(t)).$$

It has degree $p + 1$, satisfies $Q_p(a) = Q_p(b) = 0$, and has leading coefficient $(-1)^p p! (\lambda/\delta)^p$.

Proof. Write $m_p(t) = \mathbb{E}(X_t^p)$. Applying the generator associated with (4.1) to $x \mapsto x^p$ gives

$$\dot{m}_p = -\delta m_{p+1} + (\text{a linear combination of } m_1, \dots, m_p).$$

The highest moment comes from catastrophe: X_t^p is replaced by zero at rate δX_t . Solving for m_{p+1} expresses it in terms of $\dot{m}_p$ and lower moments. Along the Riccati trajectory,

$$\frac{d}{dt} = \lambda(I - a)(I - b) \frac{d}{dI},$$

so, if $m_p = Q_p(I)$,

$$m_{p+1} = -\frac{\lambda}{\delta}(I - a)(I - b)Q'_p(I) + \frac{1}{\delta}(\text{a linear combination of } Q_1, \dots, Q_p).$$

By induction, every term contains the factor $(I - a)(I - b)$. The derivative term raises the degree from $p + 1$ to $p + 2$ and multiplies the leading coefficient by $-(\lambda/\delta)(p + 1)$; the remaining terms have degree at most $p + 1$. The base case

$$Q_1(I) = J(I) = -\frac{\lambda}{\delta}(I - a)(I - b)$$

has degree two and leading coefficient $-\lambda/\delta$. The stated degree, roots and coefficient follow. $\square$

Each new moment is therefore obtained by the same operation: differentiate the previous polynomial along the Riccati flow, then add lower-order terms from births and deaths. The cases $p = 1$ and $p = 2$ are precisely (4.15) and (4.20).

The variance retains this algebraic dependence. Substituting (4.15) and (4.20) into (4.25) gives

$$\text{Var}(X_t) = \frac{2\lambda^2}{\delta^2}(I - \kappa)(I - a)(I - b) - \left[\frac{\lambda}{\delta}(I - a)(I - b)\right]^2. \quad (4.29)$$

This quartic has the common factor $(I - a)(I - b)$; the two roots are generally simple, not double. It also vanishes at $I = 1$, as it must because $X_0 = 1$ is deterministic. Along the part of the curve that has probabilistic meaning, $I \in (b, 1]$, the variance is positive for $t > 0$ and tends back to zero as $I \downarrow b$. Values on the algebraic continuation from 1 towards a are not visited by the process and need not admit a probabilistic interpretation.

The release variance is likewise a polynomial in I : substitute (4.20) and (4.22) into (4.26). Its expanded form is long but adds no new structure, so we leave it factored through the quantities already defined.

Figure 4.10 removes time from the horizontal axis. The curves are fixed algebraically, while the Riccati equation determines how quickly they are traversed. This representation becomes useful in Chapter 6, where the single-cell formulae act as age-dependent kernels in a renewal system. Since those kernels are polynomials in one monotone coordinate, they can be manipulated analytically. If a higher-order kernel is needed, proposition 4.17 supplies it by the same recursion rather than by a fresh closure argument.

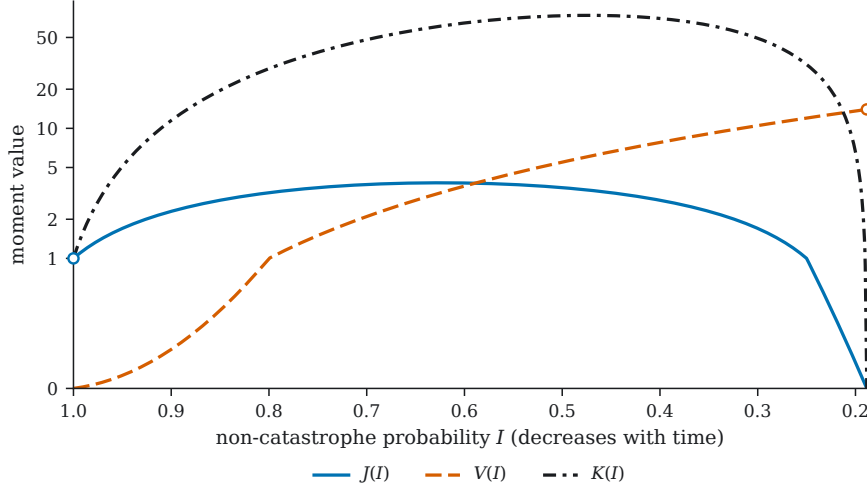


Figure 4.10: The mean load J , mean release V and second load moment K as functions of I , for $\lambda = 1$, $\mu = 0.2$ and $\delta = 0.05$. The trajectory runs from $I = 1$ towards b ; the symmetric-logarithmic ordinate keeps the three scales visible without changing their values.

4.6 Burst size and burst time

The moments just derived already tell us something about the rupture event itself. There are two quantities of interest: the burst size $\mathcal{K} = X_{\tau-}$, which is the load carried out of the cell, and the burst time τ . Neither is a marginal moment of X_t alone, but both can be reached from the functions now in hand.

4.6.1 Conditioning on rupture

The mean release $V(t)$ includes paths which have died internally and will never release anything. To obtain the mean load released by cells which have ruptured by time t , split the expectation according to the event $\{\tau \leq t\}$:

$$\mathbb{E}(W_t) = \mathbb{E}(W_t \mid \tau > t) I(t) + \mathbb{E}(W_t \mid \tau \leq t) [1 - I(t)].$$

The first conditional expectation is zero because $W_t = 0$ before rupture. Hence

$$\mathbb{E}(W_t \mid \tau \leq t) = \frac{V(t)}{1 - I(t)}.$$

Letting $t \rightarrow \infty$ conditions on eventual rupture, irrespective of when it occurs.

Corollary 4.18 (The mean burst size). *Conditional on eventual rupture,*

$$\mathbb{E}(\mathcal{K} \mid \text{burst}) = \lim_{t \rightarrow \infty} \frac{V(t)}{1 - I(t)} = \frac{V_\infty}{1 - b} = \frac{\lambda - \mu}{\delta} + \frac{1}{1 - b} = \frac{a}{a - 1}. \quad (4.30)$$

When $\mu = 0$, every realisation eventually ruptures and this conditional mean equals V_∞ .

The second moment follows in the same way. As $t \rightarrow \infty$, we have $I \rightarrow b$ and $K \rightarrow 0$, so (4.24) gives

$$\mathbb{E}(W_\infty^2) = \frac{2(\lambda - \mu)}{\delta} V_\infty + \frac{\lambda + \mu}{\delta} (1 - b) + 1.$$

Dividing by the rupture probability $1 - b$, subtracting the square of (4.30), and using (4.6) yields the variance.

Corollary 4.19 (The first two moments of the burst law). *Conditional on rupture, the burst size satisfies*

$$\mathbb{E}(\mathcal{K} \mid \text{burst}) = \frac{a}{a-1}, \quad \text{Var}(\mathcal{K} \mid \text{burst}) = \frac{a}{(a-1)^2}, \quad (4.31)$$

for every $\mu \geq 0$.

These coincide with the first two moments of a geometric distribution on $\{1, 2, \dots\}$ with ratio $1/a$, or success probability $(a-1)/a$. The agreement points towards the geometric law, but two moments do not identify a distribution. Chapter 5 derives the full burst-size law and shows that the indication is correct.

The no-death case gives a convenient check. With $\mu = 0$, we have $b = 0$, $I(\infty) = 0$, $K(\infty) = 0$ and $V_\infty = 1 + \lambda/\delta$. Equation (4.24) then gives

$$\mathbb{E}(W_t^2) \Big|_{t \rightarrow \infty} = \frac{2\lambda}{\delta} \left(1 + \frac{\lambda}{\delta}\right) + \frac{\lambda}{\delta} + 1 = \frac{2\lambda^2}{\delta^2} + \frac{3\lambda}{\delta} + 1.$$

This is $(2-s)/s^2$ for $s = \delta/(\lambda + \delta) = (a-1)/a$, the second moment of that geometric law.

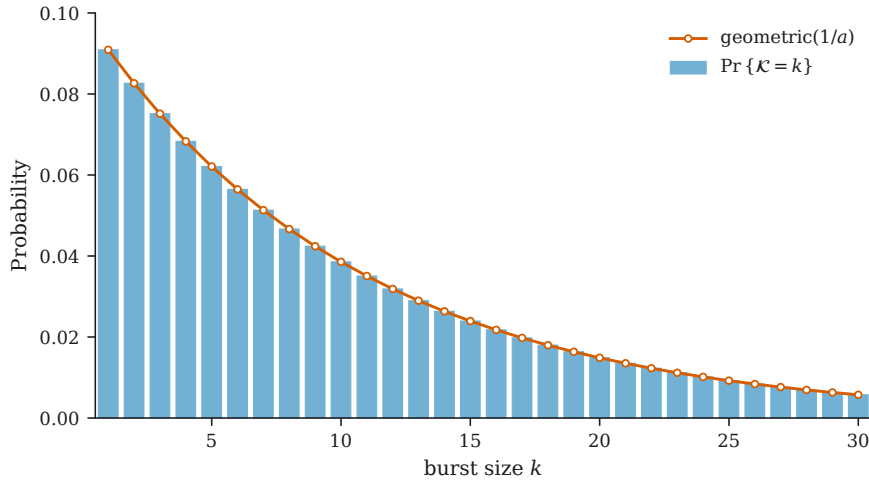


Figure 4.11: The burst-size law in the no-death case: $\mathbb{P}(\mathcal{K} = k) = (\delta/\lambda)a^{-k}$, a geometric law on $\{1, 2, \dots\}$ with ratio $1/a$. Here $\lambda = 1$, $\mu = 0$, $\delta = 0.1$ and $a = 1.1$.

Figure 4.11 previews the distribution in the case $\mu = 0$, when every path bursts and no conditioning mass is lost to internal extinction.

4.6.2 The burst time

The distribution of τ is contained in I :

$$\mathbb{P}(\tau \leq t) = \mathbb{P}(X_t = H) = 1 - I(t).$$

Its density on finite times is therefore

$$\varphi(t) = -\frac{dI}{dt} = \delta J(t). \quad (4.32)$$

The identity has a simple reading. Since any unit can trigger rupture, the instantaneous rupture density is proportional to the current mean load.

When $\mu > 0$, this density is defective:

$$\int_0^\infty \varphi(t) dt = 1 - \lim_{t \rightarrow \infty} I(t) = 1 - b < 1.$$

The missing mass b belongs to cells which die internally and consequently have $\tau = \infty$. Only in the no-death case is φ a proper density on $[0, \infty)$. The full law of τ , and its joint law with $\mathcal{K}$, are derived in Chapter 5.

4.6.3 The conditional mean and its plateau

There is a companion to the mean burst-size calculation. Although the unconditional load $J(t)$ tends to zero, the mean among cells which are still productive,

$$\frac{J(t)}{I_{\text{fix}}(t)},$$

approaches the finite value $a/(a-1)$. This is the same value as the conditional mean burst size in (4.30). The coincidence is suggestive: late-rupturing cells release the load they carry, and the conditional load itself is approaching a plateau. It is not, by itself, proof that the two limiting distributions agree. That stronger statement belongs to the quasi-stationary analysis [14, 15]; Chapter 5 establishes it by deriving the limiting conditional law.

Figure 4.12 separates the no-death and death cases. Without death, conditioning on non-catastrophe is the same as conditioning on productivity. With death, an intact cell may already be empty, and the two normalisations no longer agree.

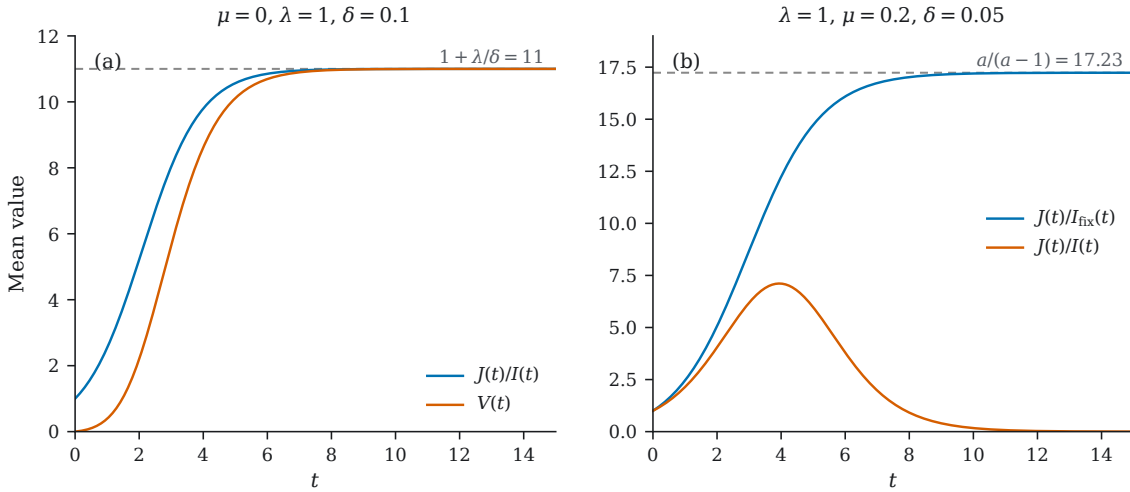


Figure 4.12: Conditional mean loads. (a) For $\mu = 0$, $\lambda = 1$ and $\delta = 0.1$, $J(t)/I(t)$ and $V(t)$ tend to $1 + \lambda/\delta = 11$. (b) For $\lambda = 1$, $\mu = 0.2$ and $\delta = 0.05$, $J(t)/I_{\text{fix}}(t)$ approaches $a/(a-1)$, whereas $J(t)/I(t)$ tends to zero because non-catastrophe also includes cells that died internally.

4.7 Summary of results

For reference, table 4.1 collects the notation, table 4.2 the principal closed forms, and table 4.3 the four differential equations from which they are obtained. Together with $I(0) = J(0) = K(0) = 1$ and $V(0) = 0$, these equations are enough to reconstruct the moment calculation.

The roots also give a compact view of parameter dependence. Figure 4.13 shows b , the probability of internal extinction before rupture, and V_∞ , the mean release over all realisations, across the (μ, δ) plane with λ fixed. Increasing the death rate makes internal extinction more likely; increasing the catastrophe rate makes it less likely. Mean release is largest when catastrophe is rare enough to permit growth, although internal death continues to compete with that growth.

Table 4.1: Notation for the birth–death–catastrophe process.

Symbol	Meaning	Established
<i>Parameters</i>		
λ	birth rate per replicative unit	(4.1)
μ	death rate per replicative unit	(4.1)
δ	catastrophe rate per replicative unit	(4.1)
<i>States and released quantities</i>		
X_t	intracellular state; numerical load is zero at H	section 4.2.4
$X_t = 0$	absorbing internal-extinction state	(4.1)
$X_t = H$	distinct absorbing catastrophe state	(4.1)
W_t	zero before rupture; $X_{\tau-}$ afterwards	(4.2)
τ	catastrophe time $\inf\{t : X_t = H\}$	(4.2)
$\mathcal{K}$	burst size $X_{\tau-}$	(4.2)
$p_n(t)$	state probability $\mathbb{P}(X_t = n)$	section 4.4.1
P_k	defective burst-size mass $\mathbb{P}(\mathcal{K} = k, \tau < \infty)$	section 4.8.1
<i>Cell-fate probabilities</i>		
$I(t)$	non-catastrophe probability $\mathbb{P}(X_t \neq H)$	(4.8)
$D(t)$	internal-extinction probability $p_0(t)$	(4.10)
$I_{\text{fix}}(t)$	productive probability $I(t) - D(t)$	(4.12)
$I_{\text{fix},k}(t)$	productive probability from k founders, $I^k - D^k$	(4.13)
<i>Moments and measures of stochasticity</i>		
$J(t)$	mean load $\mathbb{E}(X_t)$	(4.15)
$K(t)$	second load moment $\mathbb{E}(X_t^2)$	(4.19)
$V(t)$	mean release $\mathbb{E}(W_t)$	(4.22)
V_∞	late-time mean release aB/A	(4.23)
$\mathbb{E}(W_t^2)$	second release moment	(4.24)
$\text{Var}(X_t)$	$K - J^2$	(4.25)
$\text{Var}(W_t)$	$\mathbb{E}(W_t^2) - V^2$	(4.26)
$\text{Cov}(X_t, W_t)$	$-JV$	(4.27)
$\varphi(t)$	finite-time burst density $\delta J(t)$	(4.32)
<i>Parameter combinations</i>		
η	$(\lambda + \mu + \delta)/(2\lambda)$	lemma 4.4
a	$\eta + \sqrt{\eta^2 - \mu/\lambda}$, with $a > 1$	lemma 4.4
b	$\eta - \sqrt{\eta^2 - \mu/\lambda}$, with $0 \leq b < 1$	lemma 4.4
A, B	$a - 1$ and $1 - b$	(4.6)
θ	$\lambda(a - b)$	(4.6)
w	$e^{\theta t}$	(4.7)
κ	$1 + \delta/(2\lambda)$, the third root of $K(I)$	(4.20)

Table 4.2: Principal closed forms, from one founder.

Quantity	Closed form
$I(t)$	$(aB + bAw)/(B + Aw)$ (4.8)
$D(t)$	$ab(w - 1)/(aw - b)$ (4.10)
$I_{\text{fix}}(t)$	$(a - b)^2 w / [(B + Aw)(aw - b)]$ (4.12)
$J(t)$	$-\frac{\lambda}{\delta}(I - a)(I - b)$ (4.15)
$K(I)$	$\frac{2\lambda^2}{\delta^2}(I - \kappa)(I - a)(I - b)$ (4.20)
$V(t)$	$(1 - I)[1 + \frac{\lambda}{\delta}(1 - I)]$ (4.22)
V_∞	aB/A (4.23)
$\mathbb{E}(\mathcal{K} \mid \text{burst})$	$a/(a - 1)$ (4.30)

Table 4.3: The four differential relations underlying the closed forms.

Equation	Interpretation
$\frac{dI}{dt} = \lambda(I - a)(I - b)$	autonomous Riccati equation
$\frac{dI}{dt} = -\delta J$	non-catastrophe probability and mean load
$\frac{dJ}{dt} = (\lambda - \mu)J - \delta K$	mean load and second moment
$\frac{dV}{dt} = \delta K$	accumulation of mean release

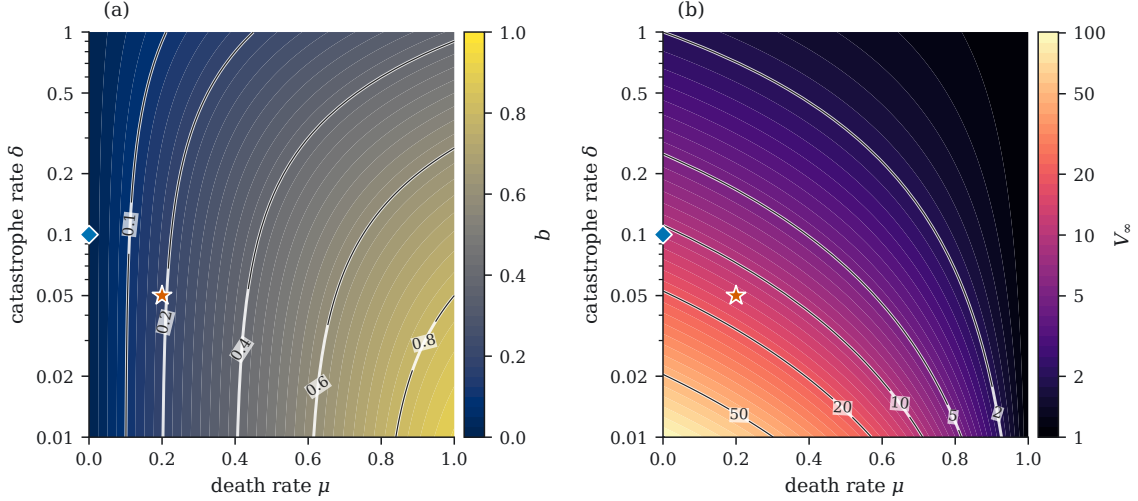


Figure 4.13: Rate landscapes at $\lambda = 1$, for $\mu \in [0, 1]$ and $\delta \in [0.01, 1]$. (a) The probability b of internal extinction before rupture. (b) The mean release V_∞ . The star marks $(\mu, \delta) = (0.2, 0.05)$, used in most figures; the diamond marks the no-death point $(0, 0.1)$. The catastrophe-rate axis is logarithmic.

4.8 Conclusion

We have described the BDC process through its cell-fate probabilities and first two moments. The resulting formulae cover the intracellular load, the released count, and the first two conditional moments of the burst size, all through the single coordinate $I(t)$.

The calculation closes because the non-catastrophe probability obeys an autonomous Riccati equation. The open moment hierarchy is still present, but it need not be truncated: it can be approached from I . The two roots of the characteristic quadratic then make the closed forms readable, and they retain their probabilistic meaning at the boundaries recorded in [section 4.A](#).

One reading takes rupture as the lysis of an infected macrophage. The formulae do not depend on that reading. They do depend on the four assumptions of [section 4.1](#): linear birth, catastrophe hazard δn , instantaneous complete release, and constant rates. Measurement of (λ, μ, δ) and the full burst law lie outside this chapter.

4.8.1 Completing the single-cell theory

Chapter 5 solves the probability generating function of the process. This yields the state probabilities $p_n(t)$, rather than only their sums I , D and I_{fix} , and identifies the quasi-stationary distribution of the intracellular load conditioned on continued productivity. It also gives the full

burst-time and burst-size laws, including the defective finite-time mass

$$P_k = \mathbb{P}(\mathcal{K} = k, \tau < \infty) = \frac{\delta}{\lambda} a^{-k}, \quad k \geq 1,$$

and the conditional laws of size given time and time given size. The several founder formulae of [section 4.3.6](#) then describe cells infected by more than one particle. Finally, the immediate-transfer model chains successive cells together by allowing one burst to found the next infection without delay.

What passes from the present chapter to that analysis is deliberately limited. We know the first two conditional burst moments, and they point towards a geometric distribution. The generating function supplies the distribution itself and explains why it agrees with the quasi-stationary load.

4.8.2 Embedding the cell in a population

Chapter 6 turns from one infected cell to many. A burst is no longer an endpoint: released particles may infect fresh cells, each of which begins its own BDC trajectory. The quantities derived here then become age-dependent kernels. The survival kernel $I_{\text{fix}}(\alpha)$ gives the probability that a cell infected α time units ago remains productive, while

$$g(\alpha) = \delta K(\alpha)$$

gives its expected rate of release. Because these functions can be written in the I -coordinate, the resulting renewal equations retain much of the analytic structure of the single-cell model.

The renewal formulation belongs to the classical theory of age-structured populations [\[16, 17\]](#). It also makes clear what is hidden by the constant-release ordinary differential equations often used in viral dynamics [\[18, 19\]](#): release from a lytic cell has an age profile and occurs in a single burst. In regimes where that timing matters, a constant-release approximation can be poor. The comparison with continuous budding release is developed in the population chapter [\[20, 21, 22\]](#).

4.A Checks and limiting cases

The main derivations assume positive birth and catastrophe rates and begin from one founder. This appendix checks the endpoints of the closed forms and then approaches the boundaries at which the two-root description degenerates.

4.A.1 Initial values

At $t = 0$, the process contains one unit and no event has occurred. Thus

$$I(0) = 1, \quad D(0) = 0, \quad I_{\text{fix}}(0) = 1, \quad J(0) = K(0) = 1, \quad V(0) = 0.$$

Setting $w = 1$ recovers each value from the corresponding closed form. The initial slopes also have a direct interpretation. Equation [\(4.14\)](#) gives

$$I'(0) = -\delta J(0) = -\delta,$$

the rate at which the founder triggers rupture. Similarly,

$$I'_{\text{fix}}(0) = -(\lambda + \mu + \delta) + \lambda = -(\mu + \delta).$$

Birth is absent from this loss rate because division leaves the cell productive; only death or rupture can remove it at the first instant.

4.A.2 Late-time values

As $t \rightarrow \infty$, $w \rightarrow \infty$, and

$$I \rightarrow b, \quad D \rightarrow b, \quad I_{\text{fix}} \rightarrow 0, \quad J \rightarrow 0, \quad K \rightarrow 0, \quad V \rightarrow V_{\infty} = \frac{aB}{A}.$$

The equality of the first two limits is not accidental. At late time, a path which has avoided rupture must have died internally; the productive probability is the vanishing difference between those two events.

4.A.3 No death

Set $\mu = 0$. From $ab = \mu/\lambda$ we have $b = 0$, while $a = (\lambda + \delta)/\lambda$. Internal extinction is then impossible: $D \equiv 0$ and $I_{\text{fix}} = I$. Every path ruptures eventually, and

$$V_{\infty} = 1 + \frac{\lambda}{\delta} = \mathbb{E}(\mathcal{K} \mid \text{burst}).$$

This is the simplest setting in which to check the geometric burst-size moments, as done in [section 4.6.1](#).

4.A.4 No catastrophe

Setting $\delta = 0$ removes rupture and returns the classical linear birth–death process. Since the main formulae contain δ in their denominators, we take the limit $\delta \downarrow 0$ in the supercritical regime $\lambda > \mu$. Then

$$a \rightarrow 1, \quad b \rightarrow \frac{\mu}{\lambda}, \quad A = a - 1 \rightarrow 0, \quad I(t) \rightarrow 1.$$

The internal-extinction probability approaches

$$D(t) \rightarrow \frac{\mu}{\lambda} \frac{w - 1}{w - \mu/\lambda}, \quad w = e^{(\lambda - \mu)t},$$

which is the standard finite-time extinction probability of the linear birth–death process from one founder. At $(\lambda, \mu, t) = (1, 0.3, 3)$, the limiting formula gives 0.27330340. Meanwhile $V_{\infty} \rightarrow \infty$: for positive but increasingly small δ , a supercritical population can grow for an increasingly long time before release.

4.A.5 Critical births and deaths

When $\lambda = \mu$, Vieta’s relation gives $ab = 1$, and hence

$$b = \frac{1}{a}.$$

At criticality, the probability of internal extinction before rupture equals the ratio of the geometric burst-size law. For $(\lambda, \mu, \delta) = (1, 1, 0.05)$, the two roots are $a = 1.25$ and $b = 0.8$. Away from criticality the equality generally disappears.

4.A.6 No births

Finally, let $\lambda \downarrow 0$. The upper root escapes to infinity and the quadratic Riccati equation becomes linear:

$$\frac{dI}{dt} = \mu - (\mu + \delta)I, \quad I(0) = 1.$$

The remaining finite root tends to

$$b \longrightarrow \frac{\mu}{\mu + \delta}.$$

This is exactly the chance that a non-replicating founder dies before it triggers rupture: the two exponential clocks divide the outcome in proportion to their rates. The two-root machinery has degenerated, but its surviving root retains the correct probabilistic meaning.

4.A.7 The second moment of the release

At rupture from state n , W_t^2 increases from zero to n^2 , and this happens at rate δn . Therefore

$$\frac{d}{dt} \mathbb{E}(W_t^2) = \delta \mathbb{E}(X_t^3). \quad (4.33)$$

The third moment has appeared, but the equation for K contains the same term. A unit change in X_t does not produce a unit change in X_t^2 .

Table 4.4: Changes in X_t^2 under the three event types.

Rate	Change in X_t	Change in X_t^2
λX_t	+1	$(X_t + 1)^2 - X_t^2 = 2X_t + 1$
μX_t	-1	$(X_t - 1)^2 - X_t^2 = -2X_t + 1$
δX_t	$-X_t$	$(X_t - X_t)^2 - X_t^2 = -X_t^2$

The entries in [table 4.4](#) give

$$\begin{aligned} \mathbb{E}(X_{t+\Delta t}^2 - X_t^2) &= \mathbb{E}(\lambda X_t \Delta t (2X_t + 1) + \mu X_t \Delta t (-2X_t + 1) + \delta X_t \Delta t (-X_t^2)) \\ &= [2(\lambda - \mu) \mathbb{E}(X_t^2) + (\lambda + \mu) \mathbb{E}(X_t) - \delta \mathbb{E}(X_t^3)] \Delta t, \end{aligned}$$

and hence

$$\frac{dK}{dt} = 2(\lambda - \mu)K + (\lambda + \mu)J - \delta \mathbb{E}(X_t^3). \quad (4.34)$$

Using [\(4.33\)](#) to remove the third moment,

$$\frac{d}{dt} \mathbb{E}(W_t^2) = 2(\lambda - \mu)K + (\lambda + \mu)J - \frac{dK}{dt}. \quad (4.35)$$

Integration from 0 to t , together with $V' = \delta K$ and $I' = -\delta J$, gives

$$\begin{aligned} \mathbb{E}(W_t^2) &= 2(\lambda - \mu) \int_0^t K \, ds + (\lambda + \mu) \int_0^t J \, ds - K(t) + K(0) \\ &= \frac{2(\lambda - \mu)}{\delta} V(t) + \frac{\lambda + \mu}{\delta} [1 - I(t)] - K(t) + K(0), \end{aligned}$$

where $K(0) = 1$. This is [\(4.24\)](#).

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